

DATA CLEANING, PREPARATION & TITLE OPTIMIZATION REPORT

1. Introduction

This project focuses on cleaning and preparing a raw marketing dataset using Microsoft Excel. The primary objective is to improve data quality by addressing issues such as missing values, duplicate records, and inconsistent formatting. Additionally, a new feature, **short_title**, was created to enhance product title readability and optimize them for search engine visibility (SEO).

2. Dataset Overview

The dataset contains product-related information including:

- Product identifiers (product_id, product_type_id)
- Product titles
- Product descriptions
- Bullet point features
- Product dimensions (e.g., product_length)

Initial exploration revealed several data quality issues, including:

- Missing values across multiple columns
- Duplicate records
- Inconsistent text formatting
- Overly long and unstructured product titles

3. Data Cleaning Process

3.1 Removal of Duplicate Records

A total of **217 duplicate records** were identified and removed using Excel's *Remove Duplicates* tool. This ensured that each product entry is unique and eliminated redundancy that could affect analysis.

3.2 Handling Missing Values

Different strategies were applied based on the nature of each column:

Product Type ID

- The product_type_id column contained missing values.
- Since this field represents a categorical identifier, missing values were filled using the **mode (most frequent value)** to preserve category distribution.

Formula used:

=IF(E2="",MODE.SNGL(\$E:\$E),E2)

Product Length

- Missing values in product_length were handled using a **group-based mean imputation**.
- The average product length was calculated for each product category using the cleaned product_type_id.

Formula used:

=IF(I2="",AVERAGEIF(\$H:\$H,H2,\$I:\$I),I2)

Description Column

- The description column had a high proportion of missing values (~56%).
- Due to this, a conservative approach was taken by filling missing entries with a contextual placeholder derived from the product title.

Formula used:

=IF(F2="", "No description available", F2)

Bullet Point Column

- Missing values (~39%) in the bullet_point column were replaced with a standardized placeholder referencing the product title to maintain usability without generating artificial content.

Formula used:

=IF(D2="", "Key features not provided - see product", D2)

3.3 Standardization of Data

- Column names were standardized using **snake_case** format (e.g., PRODUCTID → product_id)

3.4 Data Validation

The dataset was reviewed to ensure:

- No remaining duplicates
- No critical missing identifiers
- Proper formatting across numeric and text fields

- Logical consistency in values

4. Short Title Creation

◆ Objective

To generate a concise and SEO-friendly version of product titles while retaining essential product information.

Methodology

- The original product titles were processed using Excel functions such as MID, LEFT and RIGHT
- Redundant words, product codes, and unnecessary symbols were removed
- Titles were limited to approximately **30–50 characters**
- Key elements retained include:
 - Product name
 - Product type
 - Key attributes (e.g., size, quantity, compatibility)
- A combination of formula-based transformation and minor manual refinement was applied to improve readability and consistency

Examples

Original Title	Short Title
ArtzFolio Tulip Flowers Blackout Curtain...	Tulip Flowers Blackout Curtain for Door 2 pcs
Pooplu Womens Plain V Neck Half Sleeves Pack of 3 Combo Cotton Pink, Yellow, DarkPink T Shirt. Stylish, Casual Tshirts (_L)	Pooplu Womens Plain V Neck Half Sleeves - 3 color combo
Wacoal Women's Body By Wacoal Underwire Bra (38C, Red)	Wacoal Women's red underwire Bra - 38C

5. Clean Dataset Overview

After the data cleaning process:

- Duplicate records were successfully removed
- Missing values were handled appropriately based on data type

- Text fields were standardized and cleaned
- A new feature, short_title, was created

The dataset is now:

- Structured
- Consistent
- Readable
- Ready for further analysis and marketing applications

6. Conclusion

The data cleaning and preparation process significantly improved the quality and usability of the dataset. By applying appropriate techniques for handling missing values, removing duplicates, and optimizing product titles, the dataset is now well-structured and analysis-ready. The use of formula-based transformations ensures transparency, reproducibility, and adherence to best practices in data analytics.